

Design FMEA

Failure mode	Effect	Evidence/risk	Mitigation
High-rate polarization	5C capacity loss	v5 retention 0.335044	redesign chemistry/electrolyte and
Excess inactive mass	ED and mass failure	v5 mass 43.2168 g	higher-specific-energy system and pack
Thin separator robustness	safety/manufacturing risk	1 um is simulation candidate only	mechanical,

This virtual FMEA is qualitative; abuse and manufacturing tests were not performed.